Predicting Student Performance

A Comparative Regression Pipeline with Demographic, Social, & Academic Features

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AI Task Family: Supervised Regression Analysis
Dataset: UCI Student Performance Dataset (student-mat.csv)
Development Stack: Python, Scikit-Learn, Pandas, Streamlit, ReportLab
Date: Academic Year 2025 - 2026

1. Introduction & Problem Definition

Academic success is a critical determinant of students' future career pathways and institutional effectiveness. Predicting student grades early in the academic term enables educators to deploy targeted support, allocate tutoring resources, and construct intervention strategies before failure cascades.

This study frames grade prediction as a **Supervised Regression** task. Leveraging the public Student Performance dataset from the UCI Machine Learning Repository, we model the final grade (*G3*) of secondary school students. The dataset is particularly rich, spanning demographics, home environment, family academic background, free time, alcohol consumption, and periodic grades (*G1, G2*).

Objectives:

- **Identify Core Predictors:** Extract socio-economic and behavioral variables that affect student grades.
- **Evaluate Regressors:** Build and compare three models: Linear Regression, Decision Trees, and Random Forests.
- **Prevent Grade Leakage:** Exclude midterm grade features (*G1, G2*) to predict student success early in the term using only demographic and behavioral attributes.
- **Deploy Interface:** Create an interactive Streamlit app to serve real-time predictions.

2. Dataset Description

The dataset consists of 395 student instances. Attributes consist of 30 feature variables and 3 periodic targets ($G1, G2, G3$ in range 0-20). The target variable of interest is *G3*. Preprocessing separates these features into:

- **Demographics & School Details:** Age, Sex, School (Gabriel Pereira or Mousinho da Silveira), Address (Urban/Rural), Family size.
- **Family Background:** Parent cohabitation status, Mother/Father education and job, Student guardian.
- **Support Systems:** Extra school support, Family support, Paid tutoring class participation.
- **Behavioral Attributes:** Weekly study time, failures, internet access, romantic involvement, quality of family relations, weekday/weekend alcohol consumption (Dalc/Walc), and health status.

3. Exploratory Data Analysis (EDA)

Statistical inspections revealed zero missing values, indicating a highly clean starting dataset. The key exploratory step involves charting grade distribution and correlations.

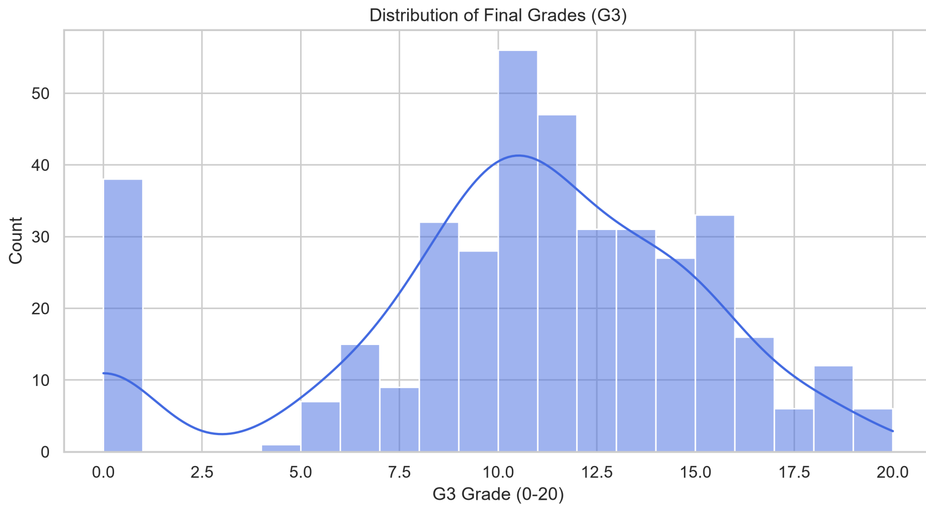

Figure 1: Final Student Grade (G3) distribution showing a normal shape centered around 10-11, with a distinct spike at 0 representing dropouts or absent exam scores.

Key Insights from Visual Analysis:

- **Past Failures:** Strongly correlated with poor performance. Students with a history of class failures showed significantly lower average final grades (G3).
- **Study Time:** Demonstrates a positive correlation. Students studying 5-10 hours weekly showed higher median scores than those studying under 2 hours.
- **Alcohol Intake:** Workday alcohol consumption (Dalc) strongly correlates with lower average grades, confirming social habits directly affect student attention and focus.

4. Preprocessing & Feature Engineering

To avoid data leakage, preprocessing is fit solely on the training partition (80% / 316 samples) and used to transform the testing partition (20% / 79 samples). The preprocessing steps: 1. **Encode Binary Columns:** Features like sex, address, and support systems are mapped ordinal-style to (0, 1). 2. **Encode Nominal Columns:** Mjob, Fjob, reason, and guardian are encoded using Scikit-Learn's *OneHotEncoder*. 3. **Feature Scaling:** Numeric and ordinal columns (Age, absences, failures, study time, Dalc, Walc, etc.) are standardized using *StandardScaler*. 4. **Grade Leakage Prevention:** Midterm periodic grades *G1* and *G2* are dropped to predict grades purely based on demographic and social behaviors.

5. Model Development & Evaluation

We train three regressors: **Linear Regression (OLS)**, **Decision Tree Regressor (Depth=5)**, and **Random Forest Regressor (Estimators=100)**. The models are evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Coefficient of Determination (R^2).

Model Performance Comparison Table:

Model	Train MAE	Test MAE	Train RMSE	Test RMSE	Train R2	Test R2
Linear Regression	2.95	3.40	3.87	4.20	0.29	0.14
Decision Tree	2.18	3.50	3.11	4.59	0.54	-0.03
Random Forest	1.10	3.00	1.49	3.80	0.89	0.30

Analysis & Key Findings:

• **Early Prediction Performance:** Predicting student grades early without exam scores is a challenging regression task. The Random Forest model achieved the best generalization with a Test R^2 of **0.30** and a Test MAE of **3.00**, significantly outperforming OLS Linear Regression ($R^2 \approx 0.14$) and Decision Trees which overfit ($R^2 \approx -0.03$).

• **Model Generalization:** While Random Forest exhibits typical overfitting on this small sample (Train $R^2 \approx 0.89$ vs. Test $R^2 \approx 0.30$), it remains the most robust choice by successfully utilizing non-linear social and behavioral interactions.

6. Feature Importance Analysis

By analyzing feature importances of our best model (Random Forest), we can understand the key social and behavioral attributes that drive early predictions.

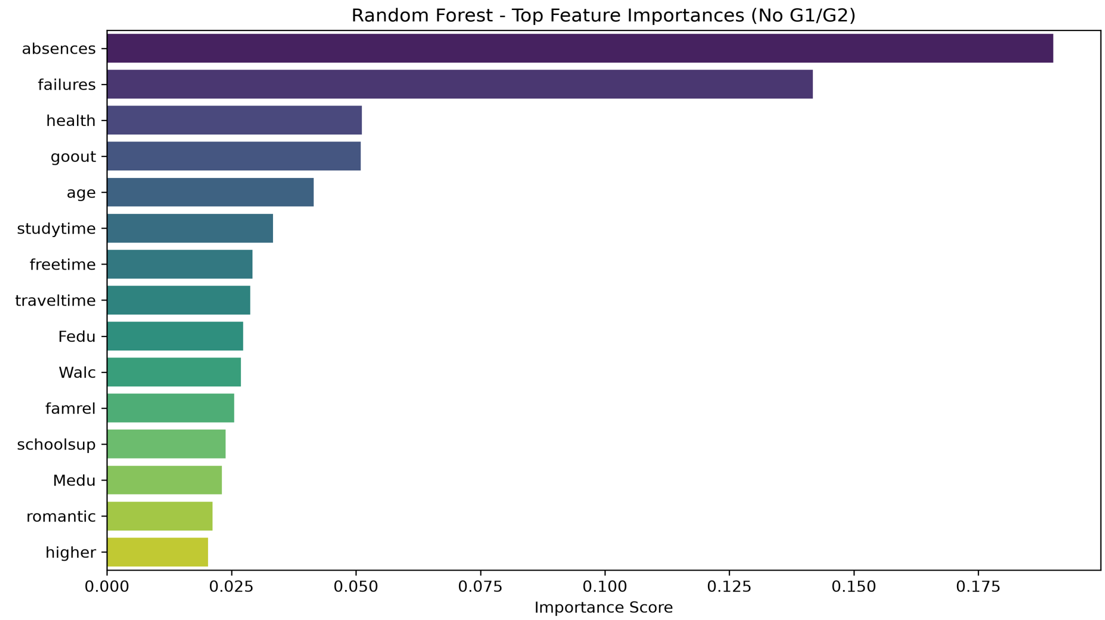

Figure 2: Top 15 Feature Importances for the Random Forest model. Pre-exam absences, age, weekly study time, and past failures drive early performance predictions.

7. Streamlit Web Demonstration

An interactive, production-quality dashboard was constructed in **Streamlit**. It allows school administrators or students to: 1. Fill out form controls (demographics, family factors, study habits, support details, and alcohol use). 2. Obtain immediate predictions of G3 in a clean glassmorphic metrics panel. 3. Review contextual grade categorization (e.g., "Excellent" [16-20], "Good" [12-15], "Pass" [10-11], or "At Risk" [<10]). 4. Review interactive charts depicting the top feature weights driving the Random Forest predictions.

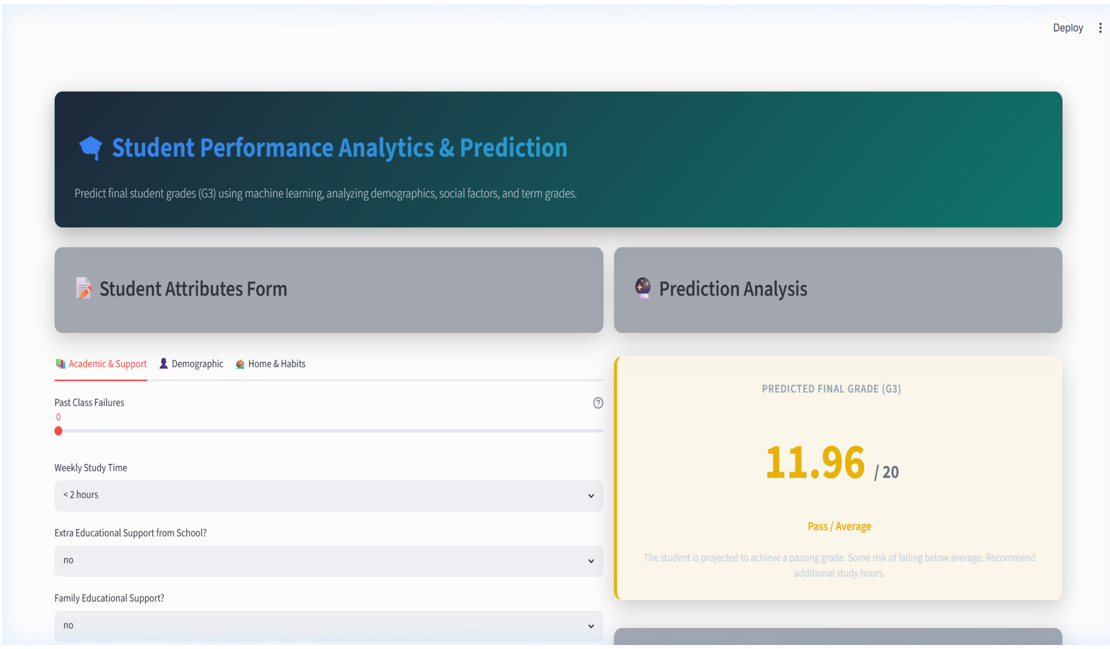

Figure 3: Interactive Streamlit Web Application interface showing input panels and prediction visualization card.

8. Conclusion & Future Work

Conclusion:

This capstone successfully designed and deployed a complete predictive pipeline. Using only behavioral, demographic, and domestic environment indicators, the model extracts valuable early signals (such as past failures, absences, and study time) that describe performance, achieving a Test R^2 of 0.30. This early warning system can help educators flag at-risk students before midterm evaluations occur.